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Meaningful time-related aspects of alerts in Clinical Decision Support Systems. A unified framework

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ABSTRACT

Alerts are a common functionality of clinical decision support systems (CDSSs). Although they have proven to be useful in clinical practice, the alert burden can lead to *alert fatigue* and significantly reduce their usability and acceptance. Based on a literature review, we propose a unified framework consisting of a set of meaningful timestamps that allows the use of state-of-the-art measures for alert burden, such as alert dwell time, alert think time, and response time. In addition, it can be used to investigate other measures that could be relevant as regards dealing with this problem. Furthermore, we provide a case study concerning three different types of alerts to which the framework was successfully applied. We consider that our framework can easily be adapted to other CDSSs and that it could be useful for dealing with alert burden measurement thus contributing to its appropriate management.

1. Introduction to *alert fatigue* and alert burden

One of the most common functionalities of Clinical Decision Support Systems (CDSSs) is that of triggering alerts to attract the clinicians' attention to those facts that may be relevant for decision-making [1–3].

Although the benefits of alerts in CDSSs are well documented [1], they tend to be overridden when they are excessive, repetitive, uninformative and/or not seamlessly adapted to the clinical workflow [2–5]. This problem, which is also known as *alert fatigue* [6], has a severe impact on both the outcomes and the acceptability of the system [5–7].

The problem of alert fatigue can have many possible causes [8–10]. Among them, the alert burden is considered one of the most relevant [7]. In this work, we focus on those time-based factors that are related to the alert burden: alert timing, cognitive overload and desensitization over time.

An inadequate *alert timing*, that is when the alert is communicated at a time that is no longer useful or affects negatively the normal clinical workflow, can lead to an increase in the perception of alert burden [2,3,10–12]. To diminish this problem, some approaches have migrated from *interruptive* alerts, where users must interrupt their workflow and accept or override the alert message, to a *non-interruptive* approach, in which the alert is displayed but does not require an immediate response. The use of interruptive alerts appears to increase alert fatigue, yet the non-interruptive approach leads to some relevant alerts being ignored and, therefore, there is no clear evidence as to which approach is the best [13–15]. The time required for a CDSS to trigger an alert should also be considered for alert timing in those systems that rely on external alert processing [16].

Another widely studied effect of an excessive alert burden is *cognitive overload*, which can be defined as the reception of a large quantity of information, along with insufficient time or cognitive resources with which to distinguish relevant from irrelevant [5]. It is commonly studied by measuring alert frequency, that is, the number of alerts received in a fixed amount of time [5,7]. This is the most common measure associated with alert burden, along with the proportion of those alerts that were overridden. However, there is no consensus about the acceptable rate to consider the problem as solved [7,17].

In the case of interruptive alerts, additional metrics have recently been proposed, such as the *alert dwell time*, which is defined as "the time elapsed from the generation of an interruptive alert to the dismissal of the alert window" [18], and the *think time* [19], defined as "the time interval between appearance of the alert and when the clinician completes all necessary actions to resolve the alert", which may include more actions (e.g., consulting with colleagues) than dwell time. These recent metrics for alert burden are useful to analyze the causes of the problem and define strategies for improvement. For example, in [20], authors suggest reviewing the usefulness of those alerts with a dwell

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Table 1
Search strings used in the bibliographic review.

Database	Search string
Scopus	TITLE-ABS-KEY((“alert fatigue” OR “alert burden”) AND (metric* OR measure* OR estimation* OR evaluation*) AND (time* OR interval* OR frequenc* OR period* OR temporal*))
PubMed	(“alert fatigue” OR “alert burden”) AND (metric* OR measure* OR estimation* OR evaluation*) AND (time* OR interval* OR frequenc* OR period* OR temporal*)

time below 2 s since they probably are not being considered and can be a cause for user frustration.

The last time-related effect of alert burden considered in this work is *desensitization over time*, which assumes that an alert is most effective when it is first noticed, and steadily becomes less effective as it is repeated over time [5]. The effect of desensitization is clear in the reduction of alert dwell time as the physicians become familiarized with a new alert [18,20], although the authors of [5] found no clear reduction in the acceptance rate over time when studying this effect on the number of overridden alerts over a long period of time.

Along with the above-mentioned works, additional reviews study the different approaches to deal with alert fatigue and/or alert burden in clinical settings. In [21], authors discuss the practices and methods they found useful to improve alerts in interruptive CDSSs. Among their proposals are measuring alert burden and effectiveness, identifying malfunctioning alerts, and developing an overall model for the approval, maintenance, and review of the system. In [12], authors review the impact of CDSSs on the workload of health professionals in general practice, concluding that the perception of an increase in consultation duration, workload, and workflow disruptions, in part caused by alert fatigue, are barriers to their usage. A review of the most usual alert metrics is performed in [7], which include the volume of alerts, dwell time, think time and others focused on the effectiveness of the alert, such as override rate or efficiency. These works coincide with the complexity of measuring alert burden and alert fatigue, concluding that there is no standard metric for it.

Instead of the high-level perspective of the available reviews in the literature, the objective of this work is to take a closer view of the relationship between time and alert burden, exploring the meaningful time-related aspects associated with alerts in CDSS. To this aim, we performed a literature review looking for studies on alert burden and/or alert fatigue, analyzed their time-related aspects, and proposed a general framework to allow the calculation of all the available metrics of alert burden. Furthermore, we included several examples of the application of the framework for both interruptive and non-interruptive alerts.

2. Materials and methods

2.1. Bibliographic research

We performed bibliographic research in Scopus and PubMed in order to identify the main time-related metrics related to alert fatigue and alert burden. To this aim, we searched for articles containing the term “alert fatigue” or “alert burden”, that also describe some measurement or estimation of these concepts, and that the metrics used are related to time. In Scopus, we limited our search to titles, abstracts and keywords to obtain the most representative studies. Additionally, we included some extra articles found during the revision of the initial results. Table 1 shows the detailed search strings employed on each database.

We obtained a total of 199 records from the search in both databases. After eliminating duplicated records (n=118), we screened the titles and abstracts looking for those works that met all the following inclusion criteria:

IC1: The paper was related to alerts or alarms.

IC2: The paper was related to the clinical setting.

We then screened the full text of the remaining papers (n=114) for an eligibility check. We excluded those papers that met any of these exclusion criteria:

EC1: The paper only mentions the topic, without using or proposing any time-related metric.

EC2: We were unable to access the full text.

At the end of the process, 77 studies were selected and considered during the development of the framework. A summary of the research process is depicted in Fig. 1.

Table 2
Classification of the bibliographic results according to the type of alert implementation discussed.

Implementation approach	N	References
Interruptive	40	[5,16,18–20,22–24,24–55]
Non-interruptive	3	[56–58]
Interruptive and non-interruptive	8	[4,7,59–64]
Alarm	16	[65–80]
Not specified	10	[17,81–89]

2.2. Analysis of bibliographic results

Fig. 2 shows the distribution of the number of selected publications per year. All the results obtained were posterior to 2008 even though we did not limit the date of the study in the search query. The highest volume of related papers was reached in 2019 and there are still recent research works related to this topic.

Regarding the approach of implementation of the alerts, most of them were interruptive alerts (n=40), only a few were focused on non-interruptive implementations (n=3) and some of them discussed both approaches (n=8). We considered as a different group those works related to alarms from electronic devices (n=16). In some cases, the implementation approach was not explicitly mentioned and we could not classify the study among any of these categories (n=10). Table 2 shows the papers included in each category.

According to the kind of clinical problem to which the alerts were implemented, a significant number of papers focused on drug-drug interaction (DDI) alerts (n=17). Many works were also focused on medication alerts others than DDI, or including it among others (n=19). Another relevant group was related to patient monitoring, mostly related to intensive care units (n=14). We considered as a separate group those works related to sepsis detection (n=6) due to their relevance. The rest of them (n=21) are general studies or discuss alerts focused on a variety of different topics, such as depression assessment, acute kidney injury prediction and venous thromboembolism prophylaxis. Table 3 shows the references included in each category.

When we studied the metrics used for estimating the alert burden, alert frequency, i.e., to count the number of alerts (or overridden alerts) within a period, was present in every study (n=77). Sometimes, the period was explicitly indicated, while in other cases it was assumed to be the duration of the study. In addition, other time-related metrics found were alert dwell time or similar approaches (n=7), alert think time or similar approaches (n=5), and the time that system needs to trigger an alert (n=2). Table 4 shows the references included in each category.

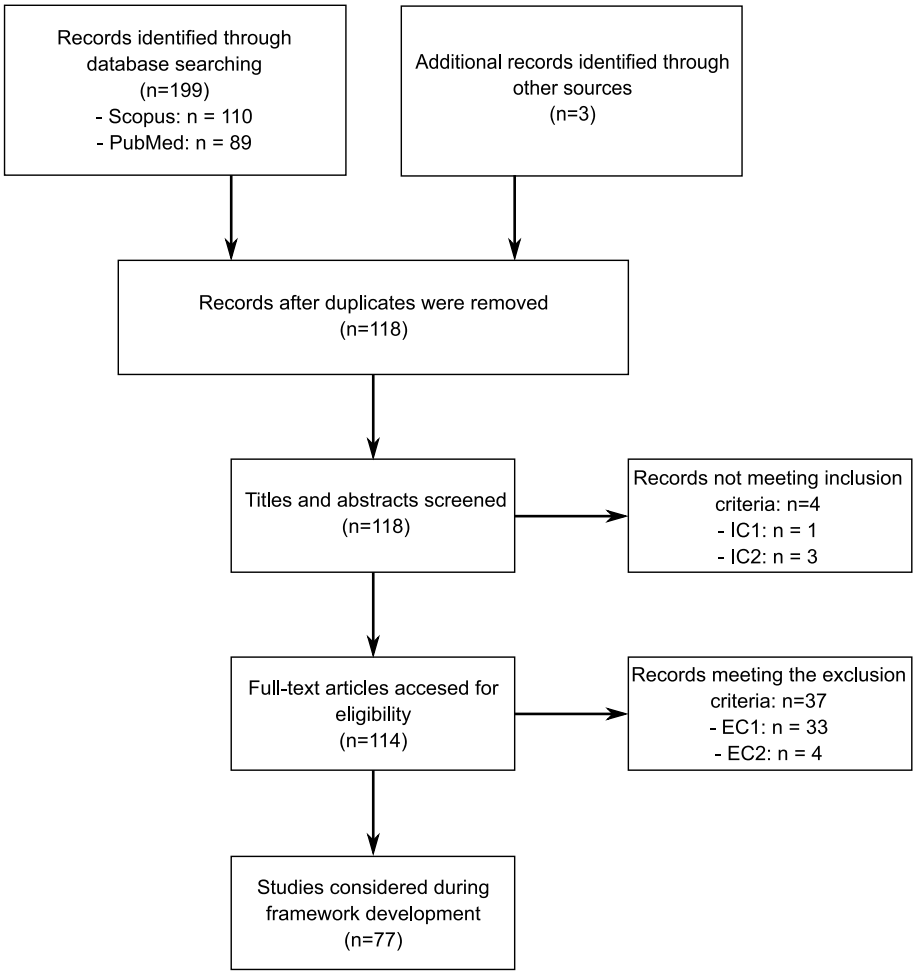


Fig. 1. Summary of the bibliographic research process.

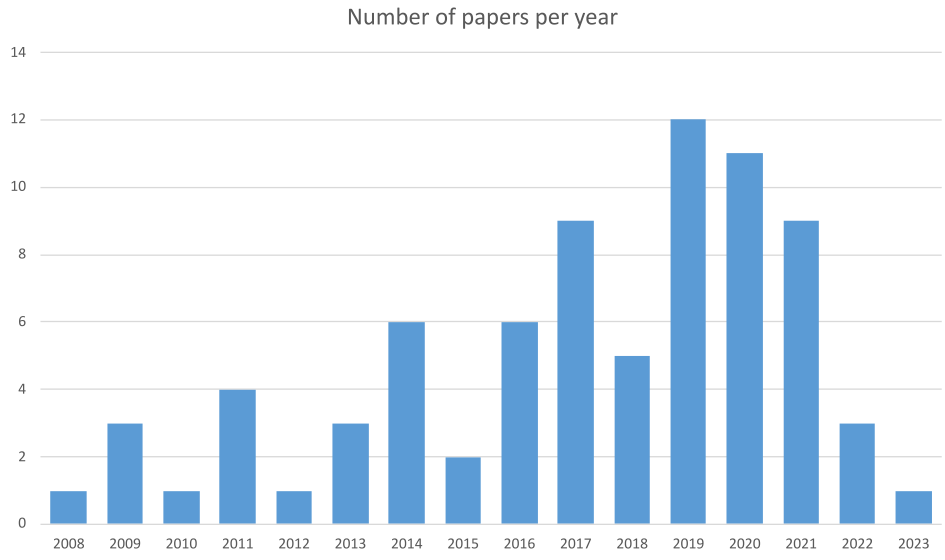


Fig. 2. Distribution of the selected papers by year.

Table 3
Classification of the bibliographic results according to the problem addressed by the alerts discussed.

Clinical problem	N	References
Medication alerts	36	
Drug-drug interaction only	17	[4,18,19,22–24,24,28,34,48–50,52,57,61,64,87]
Others	19	[25,31,33,35,37,40,45,47,51,53–56,58,60,82,85,86,88]
Patient monitoring	20	
Sepsis assessment	6	[36,41,44,66,81,84]
Others	14	[17,67–78,80]
Other alerts	21	[5,7,16,20,26,27,29,30,32,38,39,42,43,46,59,62,63,65,79,83,89]

Table 4
Classification of the bibliographic results according to the metric used to estimate alert burden.

Metric	N	References
Number of alerts (within a period)	77	[4,5,7,16–20,22–24,24–89]
Dwell time or similar	7	[7,18,20,30,41,55,63]
Think time or similar	5	[19,44,56,71,77]
System response	2	[16,84]

2.3. Meaningful time-related aspects in CDSS alerts

After a deep analysis of the time-related issues in these works, we have identified six meaningful time-related aspects of alerts that are relevant as regards dealing with alert burden.

When clinicians review an alert, they may wonder *when the events that triggered the alert occurred* [16,22,75,81,84]. This is a key question since an accurate knowledge of that instant can be decisive as regards making the best clinical decision. This instant may be significantly different from *when the alert was triggered* in some scenarios [16,65,66,68,73,75,81,84].

It is widely accepted that a timely presentation of alerts improves the acceptance of CDSSs [2,3,11]. It might, therefore, be helpful to know *when the alert was notified* so as to improve the integration of the CDSS into the clinical workflow [7,16,18–20,29,44,56,65,66,68,71,73,75,77]. Furthermore, knowing *when the user did accept/override the alert* has proven to be relevant, particularly in the case of interruptive alerts, so as to assess how much attention was paid to the alert or how much time the user took to reach a conclusion about it [7,18–20,55,56,71].

If the problem that triggered the alert has already been solved or is irrelevant for the patient's current condition, the alert will contribute to increase alert fatigue. In order to avoid this situation, it is interesting to know whether *the alert is still relevant for clinical practice* [19,22,24,38,44,56,57,62,71,77]. Furthermore, we must consider *when an alert should stop attracting the user's attention* [57,62] so as to draw the user's focus to other alerts that are more relevant for the current patient's condition.

2.4. Framework of time-related timestamps proposed for CDSS alerts

We propose a unified framework consisting of a set of meaningful timestamps to deal with the aforementioned aspects. We define these timestamps as follows:

- *Start-of-relevance timestamp (SOR)*: Instant at which the set of clinical events required to trigger an alert can be considered relevant for clinical practice.
- *Trigger timestamp (TRI)*: Instant at which the CDSS has finished the evaluation of the available clinical events and concludes that the conditions required to trigger a particular alert have been met.
- *Notification timestamp (NOT)*: Instant at which an alert is notified to the user for the first time.
- *Acknowledgment timestamp (ACK)*: Instant at which the CDSS receives feedback from the user regarding a notified alert.
- *End-of-relevance timestamp (EOR)*: Instant at which the clinical events that triggered an alert are no longer relevant as regards improving the current patient's condition.
- *Expiration timestamp (EXP)*: Instant at which the user no longer needs to focus on the alert, and it can be marked or removed to avoid attracting the user's attention.

A summary of the proposed timestamps, along with their related questions and definitions, is provided in Table 5.

2.5. Application of the framework to the measurement of alert burden

The calculation of state-of-the-art measures for alert burden can be performed by adopting our unified framework.

As shown in Section 2.2, the most used metric for alert burden is to count the number of alerts within a given time interval. Any timestamp associated with the alert might be used as a reference, although TRI and NOT would appear to be those that best match with the objective of the metric [55,64].

Alert dwell time can be calculated as the difference between ACK and NOT. The use of NOT rather than TRI or SOR makes it possible to increase the accuracy of the measure. Similarly, alert think time can be calculated as the difference between EOR and NOT.

Finally, the time required to obtain a response when the system is consulted [16] can be calculated as the difference between NOT and SOR. To achieve this, the petition required to request decision support must be considered as an additional alert-triggering event.

Fig. 3 shows the timeline of an example alert, along with the representation of each timestamp and the relevant time metrics used in literature for alert burden.

3. Use cases

In this section, we provide several examples of the use of the proposed framework on different types of alerts.

3.1. Interruptive alert example: DDI alert

Here we provide an example of how this framework may be used in the context of an interruptive DDI alert. Let us assume that the alert is triggered when a new treatment D_2 is ordered for a patient and it may interact with an ongoing treatment D_1.

We may define the value of SOR as the instant in which the user orders the treatment D_2. The system will then check the possible interactions with the ongoing treatments, find a problem with D_1 and decide to trigger the alert. This last instant will be the value of the TRI timestamp. In this setting, the values of SOR and TRI should be as close as possible.

At a certain step of the workflow, the system will decide to stop it and show the alert to the user, which will be the instant assigned to the NOT timestamp. In most cases, the value of this timestamp will be close to the two previous ones, but this may not be necessarily true in all systems. For example, the system may wait until the appropriate step of the workflow to prompt all the interruptive alerts in order to avoid distractions.

Table 5
Correspondence between meaningful time-related aspects of CDSS alerts and timestamps proposed in our framework in order to deal with them.

Time-Related Question	Timestamp	Definition
When did the events that triggered the alert occur?	Start-of-relevance timestamp (SOR)	Instant at which the clinical events are relevant for clinical practice.
When was the alert triggered?	Trigger timestamp (TRI)	Instant at which the CDSS considers that it must trigger the alert.
When was the alert notified to the user?	Notification timestamp (NOT)	Instant at which the alert is notified to the users.
When did the user accept/override the alert?	Acknowledgment timestamp (ACK)	Instant at which the CDSS receives feedback from the user.
Is the alert still relevant for clinical practice?	End-of-relevance timestamp (EOR)	Instant at which the clinical events are no longer relevant for clinical practice.
When should the alert stop attracting the user's attention?	Expiration timestamp (EXP)	Instant at which the user's attention is no longer required to be drawn to the alert.

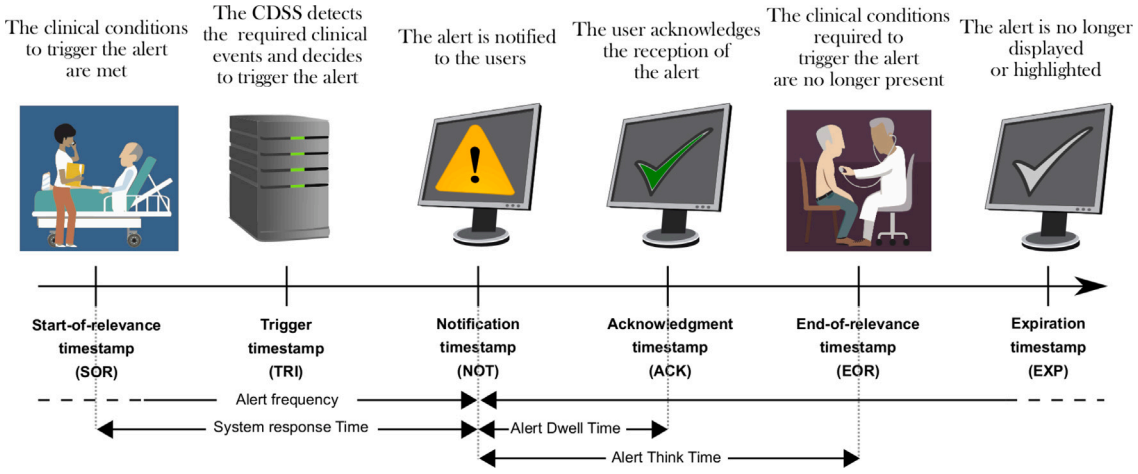


Fig. 3. Timeline view of the proposed timestamps and the intervals of interest in state-of-the-art metrics for alert burden.

Since this is an interruptive alert, the user must accept or override it in order to continue. This action will set the value of the ACK timestamp.

If the user decides to accept the alert, the order will be canceled or the previous treatment stopped. This may happen instantly if the alert is actionable, or in a short time in case the user had to perform these corrections manually. When the system detects that those changes are performed, it assigns the value to the EOR timestamp.

If the user decides to override the alert, there can be two possibilities: (1) We can assume that the alert is no longer relevant and set the value of EOR to that instant, or (2) the system can continue monitoring the possible interaction and set the value to EOR to the instant in which one of the two conflicting treatments ends.

In any case, there will be an instant at which the alert will no longer attract the attention of the user, and that will be the instant assigned to EXP. This may happen almost instantly after the alert window is closed or, for example, a mark indicating that an alert was present and overridden may linger in the application for some time in case it needs to be re-checked.

Fig. 4 summarizes the values used for each timestamp in this example.

3.2. Non-interruptive alert examples: Antimicrobial stewardship

The examples provided here are based on the Wise Antimicrobial Stewardship Program Support System (WASPSS) [90,91], a CDSS focused on antimicrobial stewardship.

Alerts in WASPSS are non-interruptive, and they are evaluated in batch. Every night, a rule-based engine checks the events registered, triggers the alerts whose conditions are met, and their timestamps are calculated. The previously triggered alerts are also revised to update those timestamps that depend on the user's clinical actions (EOR and EXP).

This particular configuration determines the definition of the TRI, NOT and ACK timestamps. Furthermore, the EXP timestamp is defined as an increment of the EOR timestamp by default. Therefore, the definitions of these timestamps in our use case are as follows:

- Trigger timestamp (TRI): Assigned by the rule-based engine the first time the alert is triggered after SOR.
- Notification timestamp (NOT): Instant of the user's first access to the list of alerts after TRI.
- Acknowledgment timestamp (ACK): Instant at which the user's feedback is received.
- Expiration timestamp (EXP): EOR + 4 days by default.

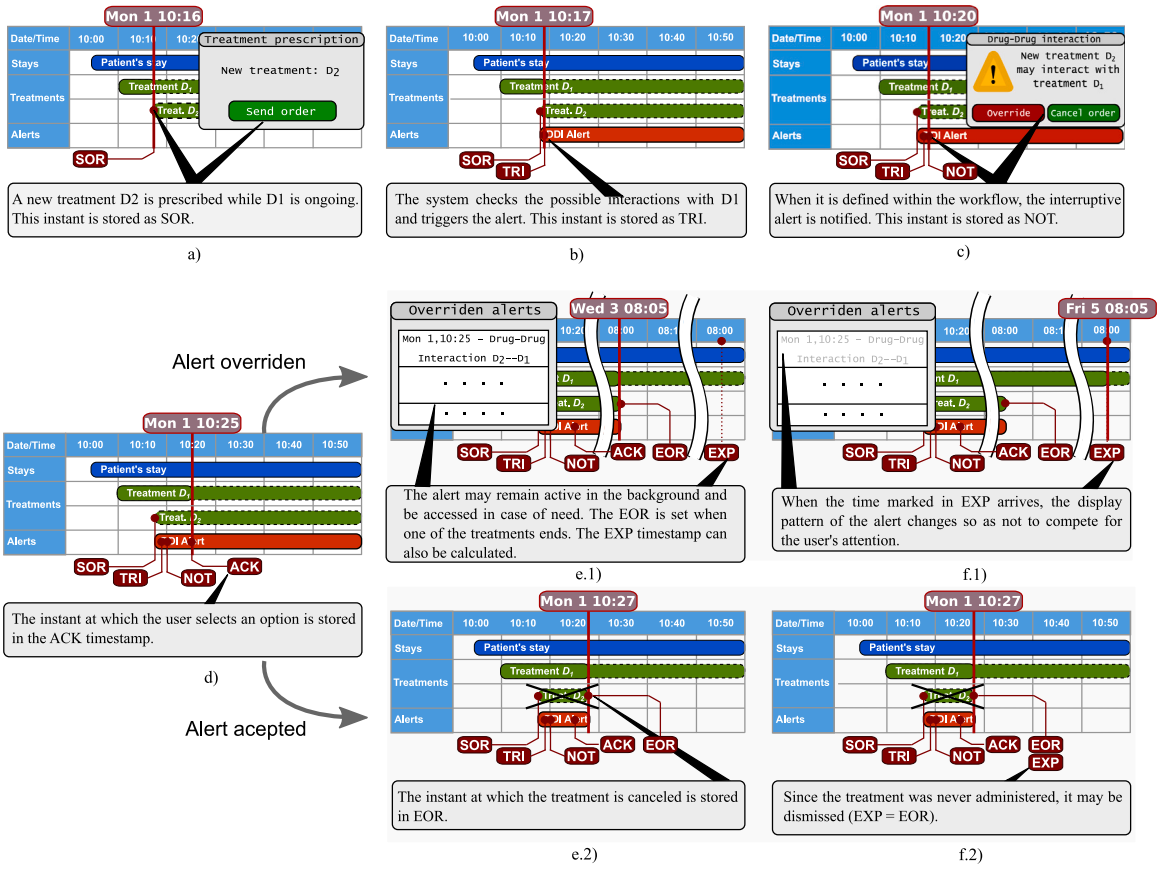


Fig. 4. Example of use of the proposed timestamps for an interruptive DDI alert. The figures depict the events that affect the calculation of the timestamps: The order of the interacting treatment (a), the detection of it by the system and trigger of the alert (b), its notification (c), and its acknowledgment by the user (d). Figures e.1 and f.1 depict the EOR and EXP calculation in case the alert is overridden, while figures e.2 and f.2 depict their calculation when the alert is accepted.

Table 6
Examples of definitions of SOR and EOR for different types of non-interruptive alerts in WASPSS.

Alert type	Related events	Possible SOR/EOR definitions
DBM	D: On-going antimicrobial treatment S: Microbiology test indicating a microorganism resistant to D	$SOR = max(D.Start, S.Start)$ $EOR = D.End$
MRM	S: Microbiology test indicating a multidrug-resistant bacteria $S_{Negative}$: Microbiology test in which the microorganism found in S is no longer detected	$SOR = S.Start$ $EOR = S_{Negative}.Start$

The assignments of the remaining timestamps (SOR and EOR) depend on the semantics of each alert. In our example, we consider two non-interruptive alerts related with antimicrobial stewardship: The Drug-Bug Mismatch alert (DBM) and the Multidrug-Resistant Microorganism alert (MRM). In WASPSS, the DBM alert is triggered when a microorganism isolated in a microbiology test is found to be resistant to an on-going patient's treatment. The MRM alert is triggered if bacteria found in a patient's sample can be classified as multidrug-resistant according to the criteria in [92].

Please note that the definitions provided here are only some of many possible options. Each institution may alter them according to the data available in their health-care systems and their clinical workflows. Figs. 5 and 6 depict some examples of these alerts, along with the relationships between their timestamps and the events that triggered the alert. A detailed definition of how SOR and EOR are calculated for these alerts is provided in Table 6.

4. Discussion and conclusions

We delved into the problem of time-related dimensions in CDSS alerts, aiming to analyze all potential scenarios in clinical practice. However, our proposal is flexible enough to allow for simplifications in some circumstances. For example, we can consider SOR, TRI and NOT as simultaneous in those CDSSs that trigger and notify alerts almost immediately after the events occur, such as those linked to Computerized Provider Order Entries (CPOEs) with interruptive DDI alerts.

As can be noticed in the use cases, our framework can be adapted to different scenarios and types of alerts. However, this implies that the calculation of some of them may vary depending on each setting. For example, the ACK in our non-interruptive alerts can only be obtained if the user confirms that the alert is valid, so there may be cases in which it is not available or must be inferred in another way. Meanwhile, it is trivial to obtain it for interruptive alerts. Nonetheless, if the appropriate mechanisms to its calculus are available, our framework allows changing the behavior of the alerts without affecting any metric used to monitor them, which is a major advantage when exploring different strategies to measure alert burden.

Our proposal has proven appropriate for the calculation of the state-of-the-art measures used for alert burden. Furthermore, it can be suitable to study other interesting time intervals. For example, the difference between EOR and SOR is the time that has elapsed between the occurrence of a clinical problem and its resolution. If we are able to identify these timestamps before and after the adoption of a CDSS,

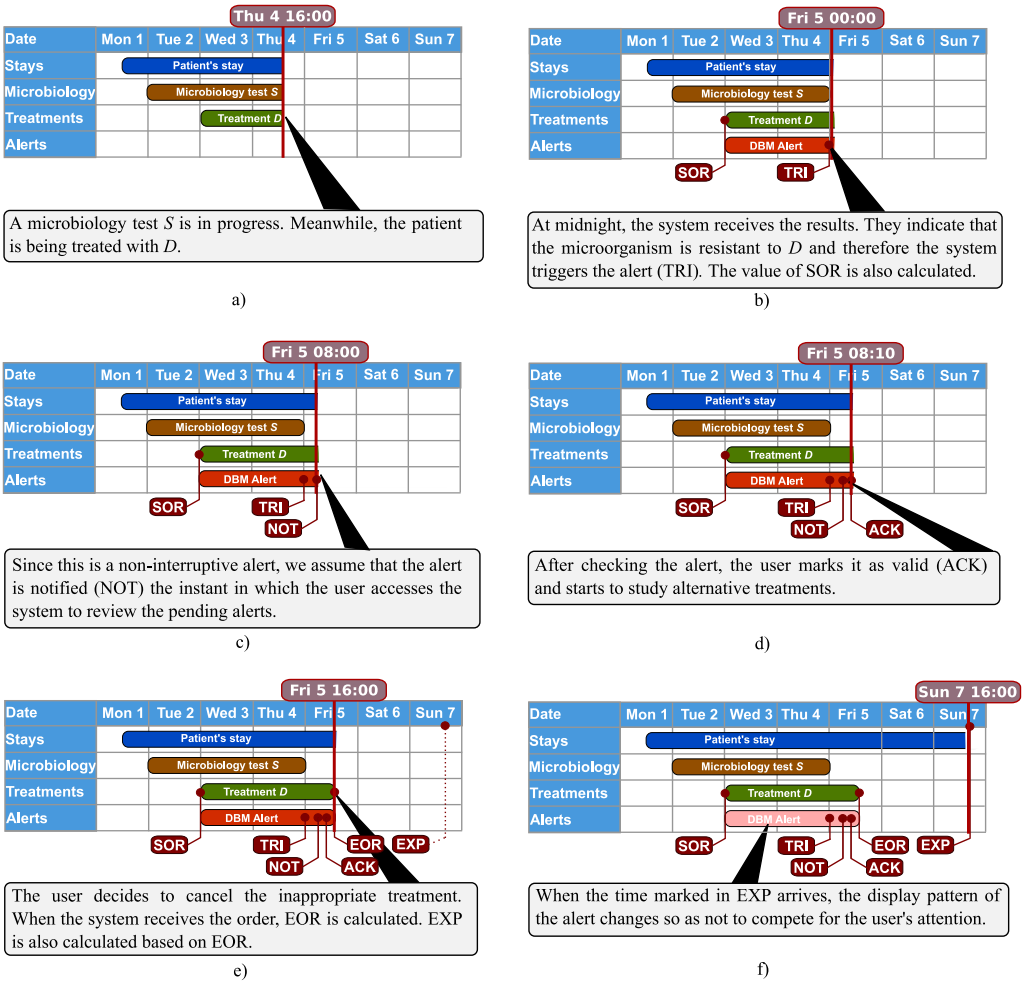


Fig. 5. Example of use of the proposed timestamps for the DBM alert. Figures a to f depict the events related to the alert before it happens (a) and the calculation of SOR and TRI (b), NOT (c), ACK (d), EOR and EXP (e) and the effect of reaching the time estimated in EXP (f). Blue bars represent the length of the patient's stay in the hospital. Green bars represent treatments, orange bars are microbiology tests, and red bars are alerts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

this may be an interesting measure of its relevance as regards the improvement of that particular clinical problem. In addition, the time that has elapsed between NOT and EXP is the total amount of time that an alert is drawing the user's attention. This could be considered a generalization of alert dwell time and could be used to design user interfaces for both interruptive and non-interruptive systems.

Furthermore, we must consider that a CDSS is usually a part of a more complex system. It may occur that the problems that triggered an alert are already being considered in another part of the workflow [93]. Our framework makes it possible to differentiate between the instant at which an alert is detected (TRI), when it is notified (NOT) and when it is no longer relevant (EOR), which allows us to explore different approaches to ensure the most suitable alert timing and avoid alert redundancy among systems.

Regarding the implementation, our proposal requires defining the relevance of events related to each triggered alert. Although it may appear to be difficult to achieve, in practice the events from which extract the timestamps are surely considered during the definition of the alert or can be calculated indirectly by using some components of the CDSS. For example, if it has an audited database, ACK will certainly be obtainable from the audit logs, and possibly TRI, NOT, SOR and EOR if all the events required for the alert are registered. This framework could also rely on other approaches that deal with the time-related aspects of information in Data Warehouses and Big Data solutions [94].

We successfully applied our framework in a CDSS focused on antimicrobial stewardship, and exemplified its usage in a common interruptive DDI alert. Owing to the different alerts tested, we consider that our approach is easy to adapt to others CDSSs and can improve the measurement of alert burden thus contributing to its appropriate management.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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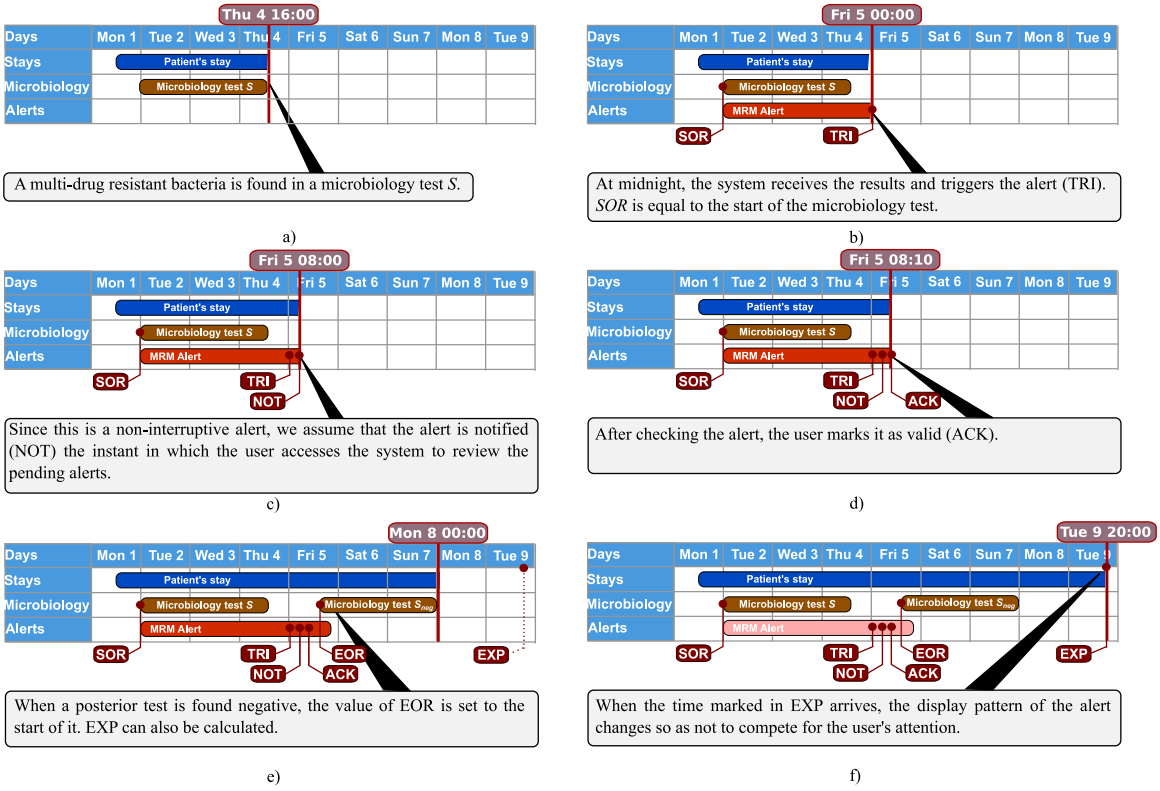


Fig. 6. Example of use of the proposed timestamps for the MRM alert. Example of use of the proposed timestamps for the DBM alert. Figures a to f depict the events related to the alert before it happens (a) and the calculation of SOR and TRI (b), NOT (c), ACK (d), EOR and EXP (e) and the effect of reaching the time estimated in EXP (f). Blue bars represent the length of the patient's stay in the hospital. Green bars represent treatments, orange bars are microbiology tests, and red bars are alerts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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